

Scalability & Future Plan

Date	6 July 2026
Team ID	xxxxxxx
Project Name	Credit Card Approval Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the Credit Card Approval Prediction System can be enhanced to support more users, improve prediction performance, strengthen security, and introduce additional features in future versions. It also presents a roadmap for expanding the application's capabilities beyond the current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No user authentication system	Anyone can access the application without login.	High
2	No database to store applicant details or prediction history	Previous predictions cannot be saved or retrieved.	Medium
3	Prediction depends on the current dataset and Random Forest model	Accuracy can be improved using larger datasets and advanced Machine Learning models.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy on a scalable cloud platform with load balancing.
2	Data Storage	No database integration	Integrate MySQL or PostgreSQL to store applicant details and prediction history.
3	Performance	Single Flask application	Optimize model loading, backend processing, and application performance for faster predictions.
4	Security	Basic input validation	Add user authentication, HTTPS, and secure data handling mechanisms.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add user authentication and account management	2–3 Months	Secure user access and personalized experience.
Phase 3	Store applicant details and prediction history using a database	3–4 Months	Enable users to access and manage previous prediction records.
Phase 4	Integrate advanced Machine Learning models and improve prediction accuracy	5–6 Months	Provide more accurate credit card approval predictions and better decision support.